

Internal Audit Certificate

Paper III — Linear-Error Clique Partitions of Split Graphs (Erdos #81)

Audit block: Block 01 — Algebraic identities (symbolic, exact)

VERDICT: PASS (12/12 identities)

Scope

Symbolic (SymPy) exact proof of the closed-form algebraic identities the paper's audit relies on: T(G) key identity (Thm 4.2), (9.12)/(9.20) coefficients, (9.19) completed square and bound, $\delta \geq 7/8$ for both parities (9.10), corridor threshold $p=2304$, μ breakpoint continuity, and the (4.5) closed forms.

Summary of evidence

- 12/12 identity checks PASS (simplify(LHS-RHS)=0 or exact rational / SOS).
- All checks symbolic over $\mathbb{Q}[p,q,d,s,\rho,\alpha]$; no floating point.

Methodology

All numeric checks use exact rational arithmetic (Python fractions) or exact linear/integer programming; symbolic identities are proved with SymPy (simplify(LHS-RHS)=0 or sum-of-squares certificate). The audit re-derives each claim independently of the manuscript's own scripts and of the Lean formalization.

Reproducibility

Results file: results/identities_results.txt

SHA-256 (results file): 7d5ea305ff7f44013184d2187d7e310efde926881cf24a4424e55667dbfe8519

Generated: 2026-07-21 16:38 UTC | Tooling: Python 3.14, SymPy 1.14, SciPy 1.17, PuLP/CBC

Auditor: internal automated audit harness (Claude Code)

This certificate attests to computational/symbolic verification of the stated finite/closed-form claims at the audited parameter ranges. It complements, and is independent of, the machine-checked Lean 4 / Mathlib formalization.